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Feature Representation, Cluster Structure and Explanation Tradeoffs in Customer Segmentation

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Abstract

Customer segmentation depends on the information used to describe each customer, yet representation is often treated as neutral. This dissertation compares a compact three-variable transaction representation with a nested twelve-variable behavioural representation for the same 11,719 RetailRocket purchasers, holding the cohort and evaluation design constant throughout. Two clustering methods are used: K Means, a standard centroid-based method, and CLASSIX, a faster method that also leaves behind its own traceable record of how each customer was grouped (its *provenance*). A separate technique, ExKMC, is used only afterwards to approximate the K Means groupings with simple rules.

Changing the representation produces markedly different customer groupings. Comparing the two independently chosen K Means partitions to each other gives an adjusted Rand index (a standard 0-to-1 agreement score between two groupings of the same people) of only 0.014; the two CLASSIX partitions agree at 0.068. Both scores are close to what unrelated random groupings would give by chance. At the same time, refitting each representation on repeated 80 per cent subsamples reproduces its own grouping with mean agreement above 0.964. A representation can therefore be highly self-consistent while still disagreeing sharply with the other representation of the same customers.

The explanation tradeoff also changes with representation. Approximating the K Means groupings with a four-rule decision tree reaches 0.9448 held-out accuracy in the compact space and 0.9650 in the richer space; CLASSIX's own traceable record grows from 131 to 559 intermediate groups when the richer representation is used. Passing a documented, size- and coverage-based audit of whether a cluster could realistically be targeted with a simple business rule, only one of the four richer-representation K Means clusters succeeds, and no compact-representation or CLASSIX cluster does.

Representation therefore changes observable cluster structure and the conditions needed to explain it. Richer behavioural information does not automatically improve internal cluster quality or produce deployment-ready segments; its demonstrated value here is limited to descriptive and queryable structure, pending prospective evidence that such segments respond differently to any actual business intervention.

Declaration

This report is my original work unless referenced clearly to the contrary. No portion of the work referred to in this report has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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AI Statement

Artificial intelligence tools were used as assistants for code drafting, data processing, numerical checks, source discovery and English language polishing. The author determined the research design, verified all sources and results, reviewed every substantive claim and accepts responsibility for the report. Details relevant to reproducibility are recorded in the accompanying materials.

Chapter 1

Introduction

1.1 Research background

Customer segmentation converts behavioural records into groups that can be inspected, compared and queried. A clustering algorithm never observes a customer directly. It receives a numeric vector assembled from selected events, time windows and aggregation rules — this vector, and the set of rules used to build it, is what this dissertation calls a customer *representation*. A compact purchase summary emphasises transaction timing and repetition. A richer representation also records browsing volume, conversion ratios, temporal engagement and category breadth. Because a clustering algorithm only ever sees the representation, not the customer, changing the representation changes the distances the algorithm measures between the same people, and can make different group structures visible or invisible.

This dissertation studies that effect for one specific, bounded population: visitors to an e-commerce platform who completed at least one purchase, drawn from the RetailRocket event log described in Chapter 3. The general problem — that a clustering algorithm’s output depends on how customers are encoded, not only on which customers are studied — is not specific to this dataset. The empirical evidence that follows, however, is: it demonstrates the effect for this purchasing cohort under one feature contract, and does not claim that the same magnitude of effect, or even the same direction of effect, would be observed for non-purchasing visitors, a different platform, or a different set of added variables.

Representation is often evaluated only indirectly, through the score of whatever clustering the analyst finally settles on. That approach cannot separate the effect of the information choice from the effect of algorithm behaviour, because the cohort, the parameter grid and the validation rule are usually all changing at the same time. It can also reward a clustering with a high silhouette score (an internal measure of how compact and well

separated clusters are, defined precisely in §3.2.3) that isolates a handful of unusual observations while leaving almost every other customer in a single undivided group — a partition that scores well but is operationally useless. *Clusterability* research — the study of whether, and by which criterion, a dataset can be said to contain genuine recoverable group structure at all — shows that this question depends on the definition adopted and on the data’s own conditions rather than on one universal test (Ackerman and Ben-David, 2009; Adolfsson, Ackerman and Brownstein, 2019). Representation must therefore be examined as part of the empirical model being fitted to customer event logs, not treated as a preliminary, inconsequential step that precedes the ”real” modelling choices.

Explanation introduces a second, independent source of ambiguity. CLASSIX, one of the two clustering methods used in this dissertation, produces *provenance*: a traceable, step-by-step record of which points were grouped together, in what order, and by which rule, so that a customer’s final cluster label can be traced back through the intermediate groups that produced it (Chen and Güttel, 2024). A decision tree fitted afterwards to approximate K Means’s cluster labels — the approach taken by ExKMC in this dissertation — instead provides a *post hoc* approximation, whose accuracy against the original labels (its *fidelity*) and whose size (its rule count) can both be measured (Dasgupta et al., 2020). These two kinds of explanation answer different questions about different objects. A short, high-fidelity tree can closely reproduce a reference partition without revealing anything about how that partition was originally computed. A detailed intrinsic trace can account for every single assignment while still being too large to summarise as a handful of business rules. Treating these two kinds of evidence as interchangeable, or combining them into one explainability score, would hide exactly the effect this dissertation is designed to detect: that representation changes both the cluster structure itself and the burden of explaining it.

1.2 Research gap

Two gaps in the existing literature motivate this study, and both are developed further in Chapter 2. First, representation is usually held fixed, or varied only incidentally, while attention is given to algorithm choice or to the final clustering score; the joint, controlled effect of representation on both structure and stability, for the same customers under matched grids and constraints, is rarely isolated directly. Second, explainable clustering methods are frequently discussed as a single category, when they in fact occupy at least three distinct design positions: *intrinsic* methods (such as CLASSIX) that expose the computation producing their own partition; *explainability-first* methods that build a simple description into the clustering objective itself; and *post hoc* methods (such as ExKMC) that approximate an already-fitted reference partition afterwards. These

three positions are not interchangeable, and conflating them risks attributing a property demonstrated for one (for example, ExKMC’s measured fidelity) to a different kind of evidence (for example, CLASSIX’s own provenance).

1.3 Research questions

This dissertation answers five research questions, each of which corresponds directly to one part of the experimental programme carried out in Chapters 3 and 4:

- RQ1. How much do independently selected K Means and CLASSIX partitions of the same purchasers agree with each other when the customers are described by a compact transaction representation compared with a richer behavioural representation?
- RQ2. Does high within-representation resampling stability (repeatability of a fitted configuration when refit on repeated subsamples of the same representation) coexist with low cross-representation agreement, and what does that combination imply about representation as a modelling choice rather than a preprocessing step?
- RQ3. How does representation change the two available explanation channels — CLASSIX’s own intrinsic provenance and ExKMC’s post hoc, held-out fidelity — and the computational or complexity profile of each?
- RQ4. Which selected clusters, if any, are operationally addressable (large enough, sufficiently data-complete, and distinctive enough on named features to be targeted with a simple query), and how does this depend on representation and clustering method?
- RQ5. Do the observed representation effects persist under sensitivity checks — feature deletion, PCA, small numerical perturbations, alternative session-window definitions, and a permutation control — or are they artefacts of one specific parameter choice?

1.4 Research purpose, significance and contribution

The purpose of this dissertation is to answer these five questions for one matched pair of representations of the same 11,719 RetailRocket purchasers. The compact representation contains three variables describing purchase recency and frequency. The richer representation retains those three variables unchanged and adds nine further variables describing activity volume, conversion, temporal engagement and category behaviour. Both representations use the same event snapshot, the same customer order, the same transformations, the same algorithm grids, the same deployment constraints

and the same resampling seeds. This design compares the effect of adding behavioural information within one recorded population; it does not compare different cohorts, different snapshots, or different algorithms’ default settings against each other.

Holding the cohort and the evaluation rules fixed in this way is what makes the comparison informative: it isolates the effect of representation from changes in population or in model selection that would otherwise be entangled with it. The analysis also keeps CLASSIX’s intrinsic provenance and ExKMC’s post hoc approximation strictly separate, because — as argued above — they explain different objects and are not comparable on one shared scale. A supporting mathematical section (§3.2) restates why CLASSIX’s projection-based aggregation search can safely skip some full-distance comparisons (its *projection filtering* step), and derives a local, sufficient condition under which that aggregation search provably reaches the same result before and after a small perturbation of the representation.

This dissertation makes five contributions, corresponding to RQ1–RQ5 above:

1. a paired comparison of compact and rich representations of the same customers, rather than a comparison across unrelated cohorts or studies;
2. a joint clustering-structure and resampling-stability comparison for two different algorithm families (K Means and CLASSIX) under matched grids and deployment constraints;
3. an explicit separation of CLASSIX’s intrinsic provenance from ExKMC’s post hoc fidelity as two distinct, non-interchangeable kinds of explanation evidence;
4. a multi-part robustness and sensitivity analysis (feature deletion, PCA, numerical perturbation, alternative session windows, and a permutation control) that tests which parts of the representation effect survive specific, named alternative explanations;
5. a bounded mathematical analysis of CLASSIX’s projection-filtering step and of the conditions under which its distance-merging aggregation result is provably stable to small perturbations.

None of these contributions supports, and none is intended to support, a claim that either representation or either clustering method is universally superior; each is scoped to what the corresponding evidence in Chapter 4 actually establishes.

1.5 Research content and headline findings

The empirical programme begins with nine externally labelled benchmark datasets, used to check that the four clustering implementations used later — K Means, CLASSIX, and, for this benchmark only, two comparison methods, DBSCAN (a density-based clustering

algorithm) and Ward’s method (an agglomerative hierarchical clustering algorithm) — and the label-free selection procedure behave sensibly across a range of known cluster shapes before either is applied to the RetailRocket data. DBSCAN and Ward’s method appear only in this supporting benchmark; they are not used anywhere in the main RetailRocket analysis. The main analysis then applies K Means and CLASSIX to both RetailRocket representations and compares cluster counts, internal indices, customer assignments and fixed-parameter overlap-resampling stability (the repeatability of a fitted configuration across repeated 80 per cent subsamples). CLASSIX provenance and held-out ExKMC trees provide the two explanation channels described above. Feature deletion, PCA, numerical perturbation, alternative session windows and a permutation control examine sensitivity. A supporting mathematical analysis (§3.2.5–§3.2.6) states the scope of CLASSIX’s projection filtering, its distance complexity, the conditions for deterministic execution, and a local sufficient stability condition.

The headline empirical result concerns cluster agreement. The two independently selected K Means partitions — one fitted on the compact representation, one on the rich representation, of the *same* 11,719 customers — agree with each other at an adjusted Rand index (ARI) of 0.014; the two CLASSIX partitions agree at 0.068. The adjusted Rand index measures agreement between two partitions of the same objects: it equals 1 for identical partitions and is corrected so that pairing labels at random scores approximately 0. Critically, there is no external ground-truth segmentation of these customers to compare against — the “two partitions” being compared throughout this dissertation are always the compact-space and rich-space clusterings of the same people, never a clustering against a known correct answer. An ARI of 0.014 or 0.068 therefore does not mean either clustering is 1.4% or 6.8% “accurate”; it means the two representations of the same customers produce partitions that are little more similar to each other than two unrelated random groupings would be, despite both being computed from the same underlying event history. Compact representations give higher silhouette values, while the rich K Means partition is more balanced across its four clusters and the rich CLASSIX partition requires more internal aggregation groups (a measure of computational provenance, not necessarily of human interpretability). At the common four-leaf ExKMC budget, held-out fidelity is 0.9448 for the compact K Means approximation and 0.9650 for the rich one. Of the resulting clusters, only one — a rich-representation K Means cluster — passes the documented addressability audit. Together, these findings show that changing the represented information changes both the observed grouping and the burden of explaining it, for this cohort and this feature contract; they do not establish that one representation is universally preferable, nor that any resulting cluster has realised business value.

Chapter 2

Literature Review

This chapter is organised around the conceptual problem the dissertation addresses, rather than as a list of individually summarised papers. §2.1 discusses how representation is treated in customer segmentation research. §2.2 discusses cluster structure and validation. §2.3 distinguishes three design positions in explainable clustering. §2.4 discusses how those methods measure and control explanation complexity. §2.5 states the resulting research gap and positions this dissertation within it.

2.1 Representation in customer segmentation

Research on customer segmentation often treats representation as a *neutral input* to clustering — that is, it implicitly assumes that the particular choice of which behavioural signals to encode is a preliminary, largely inconsequential step, and that any reasonable encoding of the same customers would support broadly the same substantive conclusions. That assumption is difficult to defend for *event logs* — timestamped records of individual visitor actions such as page views, cart additions and transactions, of the kind analysed in this dissertation (see §3.1.1). A customer can be described by a handful of purchase summaries, by the full raw sequence of recorded actions, or by a richer set of temporal, conversion and category signals computed from that sequence; each choice changes the distances the clustering algorithm observes between the same people. Clickstream research describes raw action sequences as large, variable and noisy, and instead frames useful segments as subsets recovered through constraints on *derived* attributes (Dextras-Romagnino and Munzner, 2019). Work on object-centred event logs likewise converts raw traces into vector profiles before any clustering or pattern analysis is attempted (Faria Junior et al., 2023). Customer behaviour studies construct features for segmentation and subsequent decision tasks, but their application-specific choices do not by themselves establish a universally correct representation (Abdi and Abolmakarem,

2019). Representation is therefore better understood as part of the empirical model being fitted to customer event logs, rather than as a preliminary clerical step that precedes the “real” modelling decisions.

2.2 Cluster structure and validation

Clustering does not reveal a representation-independent truth about a population. Clusterability theory supplies several formally incompatible notions of when data possess recoverable structure at all (Ackerman and Ben-David, 2009), and empirical comparisons show that clusterability measures themselves depend on distributional conditions, dimensionality and the type of structure sought (Adolfsson, Ackerman and Brownstein, 2019). A high internal validity score therefore supports a partition *within* the feature space it was computed in — it is not evidence of an intrinsic segmentation that exists independently of that space. Dimensionality reduction retains this dependence rather than removing it: it replaces named variables with a new, rotated geometry, and in doing so can obscure the link between a resulting cluster and the original, business-meaningful attributes (Bertsimas, Orfanoudaki and Wiberg, 2021). A paired comparison on the same customers — the design adopted in this dissertation — is accordingly more informative about the effect of representation than comparing results drawn from unrelated populations or studies, because it holds the population fixed and varies only the representation.

2.3 Explainable clustering

Explainable clustering methods are often discussed as a single category, but they occupy at least three distinct design positions, which must not be conflated.

Intrinsic methods expose the computation that produces a partition, as part of producing it. CLASSIX sorts observations along a principal direction (the direction of maximum variance in the standardised data), aggregates nearby points into groups starting from a sequence of starting points, and merges the resulting groups according to a stated rule (Chen and Güttel, 2024); its public group assignments, starting points and final labels together support a step-by-step trace from any one observation to its final cluster.

Explainability-first methods instead build a simple description directly into the clustering objective, so that simplicity is optimised for jointly with cluster quality rather than added afterwards. ICOT constructs tree-shaped partitions through combinatorial optimisation, while XClusters balances the usual clustering distortion objective against the size and accuracy of an accompanying tree during the formation of the clusters themselves

(Bertsimas, Orfanoudaki and Wiberg, 2021; Hwang and Whang, 2023). **Post hoc** methods begin from an already-fitted reference partition and approximate it afterwards with rules or a tree, without influencing how that reference partition was formed; ExKMC, used later in this dissertation, is one such method.

These three positions answer different questions about different objects, and their outputs are not interchangeable. Provenance from an intrinsic algorithm answers how its *own* grouping arose. A post hoc tree answers how closely a set of axis-aligned rules can reproduce *another*, already-fixed partition. Neither answer can be converted into the other without changing what is being explained — a point developed further for CLASSIX and ExKMC specifically in §2.5 and in Chapter 4.

Table 2.1 Comparison of explanation designs in the reviewed literature. *Timing* records whether the explanation is produced intrinsically as clustering proceeds, jointly with the clustering objective, or post hoc after clustering is already complete. *Complexity control* records the specific setting or budget through which each method limits how elaborate its explanation can become.

Method	Target	Timing	Form	Complexity control	Status
CLASSIX	Own partition	Intrinsic	Groups and starting points	Radius and minimum size	Journal
IMM	K Means objective	Tree construction	Threshold tree	One leaf per cluster	Conference
ExKMC	K Means labels	Post hoc	Expanded tree	Leaf budget	Preprint
CREAM	Fitted partition	Post hoc	Cubes and tree rules	Hypercube tuning parameters (coverage/purity thresholds)	Conference
ICOT	Own partition	Joint	Optimised tree	Validation objective and tree size	Journal
XClusters	Joint objective	Joint	Decision tree	Distortion and node tradeoff	Conference

2.4 Explanation complexity and fidelity

Decision trees make the tradeoff between concise expression and partition quality explicit. IMM uses threshold cuts to construct a small tree and analyses the additional K Means or K Medians cost that restriction causes (Dasgupta et al., 2020). Theoretical work shows that this kind of simplicity can impose an unavoidable loss in clustering quality, and that the size of that loss depends on the specific clustering problem, the number of clusters and the dimensionality (Laber and Murtinho, 2021). Related work asks whether a fixed clustering can be made explainable after removing a limited number of outlying observations, which further demonstrates that an apparently simple tree can obtain its simplicity by quietly changing the population being explained, rather than by finding a genuinely simpler description of the original population (Bandyapadhyay et al., 2022). CREAM constructs hypercubic approximations through decision-tree induction and tunes the resulting approximation directly, extending the same concern to a second reference algorithm (Sabbatini and Calegari, 2023). Together, these methods make complexity measurable — through leaf count, tree depth, the number of conditions per rule, and the number of features used — but that measurability does not by itself establish that a person will understand, trust or act on the resulting explanation.

Theory and software maturity vary across these methods in ways that matter for how their results can be used. IMM and the price-of-explainability analyses above provide mathematical guarantees on how much clustering quality can be lost by imposing a simple tree, under precisely defined objective functions. CLASSIX, by contrast, supplies a maintained software implementation and public explanation attributes for its own fitted partition, but it does not provide the same kind of quality-loss guarantee. CREAM, ICOT and XClusters each broaden the space of allowed explanations or add joint optimisation of size and accuracy, but their formal guarantees and their computational cost concern different targets from either CLASSIX or IMM. The practical implication is that queryable rules, a bounded loss in clustering quality, and native intrinsic provenance are three separate, complementary properties of an explanation method — not three points on one common "explainability" scale that would let these methods be ranked against each other.

ExKMC expands a decision tree beyond one leaf per K Means cluster, so that the analyst can test whether allowing more terminal nodes (a larger *leaf budget*) improves agreement with a fixed reference partition. Its only verified source is an arXiv preprint rather than a peer-reviewed publication; it documents the implemented method but does not itself provide independent evidence of general effectiveness (Frost, Moshkovitz and Rashtchian, 2020). *Fidelity*, as used throughout this dissertation, means the proportion of exact label agreements between the tree's predictions and a set of held-out K Means assignments — it is neither an internal clustering quality metric nor evidence that the underlying K Means

partition is substantively correct, only a measure of how closely the tree reproduces it. More general explainable-AI research finds that fidelity measures can disagree with each other even when a fully transparent reference model is available, so the specific target being approximated and the specific calculation used must always be stated explicitly rather than assumed (Miró-Nicolau, Jaume-i-Capó and Moyà-Alcover, 2025).

Operational usefulness introduces a further, distinct boundary. A decision tree or a set of threshold conditions can be directly queried in a database, while an intrinsic trace can instead identify representative points and merge relationships. *Queryability* can be observed through the number of conditions in a rule, whether those conditions use named (business-meaningful) features, how many customers the rule covers, and how closely it agrees with a reference partition — but none of these observable properties by itself establishes improved revenue, a more effective intervention, or genuine human comprehension. Segmentifier illustrates why recorded constraints and provenance can support the interactive refinement and export of behavioural segments, but its design study does not attempt to validate the commercial effect of the resulting clustering model (Dextras-Romagnino and Munzner, 2019). Equally, a theoretical guarantee on objective cost does not by itself measure whether a resulting segment is large enough, sufficiently distinct, or covered by reliable enough attributes to be acted upon in practice — this last question is addressed directly, for the RetailRocket clusters, by the addressability audit introduced in §3.2.3 and reported in §4.3.

2.5 Research gap and positioning of this dissertation

Two gaps follow from the discussion above, and both motivate the design of this dissertation. First, the effect of representation on clustering is rarely isolated directly: it is usually varied incidentally alongside algorithm choice, cohort, or validation rule, rather than compared for the same customers under matched grids and constraints. Second, and specifically for the two explanation methods used later in this dissertation, CLASSIX provenance and ExKMC fidelity answer genuinely different questions and must not be interpreted as one common explainability score: CLASSIX provenance is exact evidence about how CLASSIX’s *own* fitted partition was produced, whereas ExKMC fidelity is evidence about how well a tree can imitate a *separately fitted* K Means partition. A high ExKMC fidelity says nothing about how faithfully CLASSIX’s aggregation groups trace their own clusters, and a large or small number of CLASSIX aggregation groups says nothing about how simple a post hoc tree for K Means on the same data would need to be. In line with these observations, the present study compares compact transaction and rich behavioural representations for the same RetailRocket purchasers under matched clustering and evaluation conditions, and relates the resulting differences in cluster

structure to CLASSIX provenance, ExKMC fidelity and auditable addressability as three separate, non-interchangeable strands of evidence.

Chapter 3

Data and Methodology

This chapter explains how the RetailRocket event data are converted into two customer representations and evaluated under matched analytical conditions. It also distinguishes the mathematical properties of CLASSIX from the empirical robustness checks conducted in this study.

3.1 Research design

The research design compares two descriptions of the same purchasers. The compact representation contains three variables that summarise purchase timing and frequency. The rich representation contains twelve variables. It retains the same three purchase variables and adds nine measures of activity, conversion, temporal engagement and category behaviour. Both representations are constructed from the same event history for the same 11,719 customers. Their comparison therefore examines how the additional behavioural variables alter the geometry presented to the clustering algorithms.

3.1.1 Data source

The empirical comparison uses the RetailRocket recommender system dataset, an event log released under the Creative Commons Attribution NonCommercial ShareAlike licence (Retailrocket, no date). The data used in the analysis contain 2,756,101 event rows, two item property files with 20,275,902 rows in total, and a category tree with 1,669 nodes. Events record an anonymous visitor, time, item and action type. Transaction identifiers are present on purchase events, but price, expenditure and revenue are absent. Event frequency is not interpreted as monetary value, and purchase rows are not described as product quantities.

The study population contains every visitor with at least one transaction event. Applying this rule to the event table yields 11,719 purchasers and 230,678 event rows for those purchasers. Both representations use these same customers in the same order. The observation snapshot is one day after the final recorded event, which gives 19 September 2015 at 02.59.47 UTC. Purchase recency is measured from this snapshot rather than from each customer's final browsing action. Stable ordering by visitor, timestamp and original row position resolves equal timestamps without making an arbitrary event sequence depend on file reading behaviour.

3.1.2 Feature engineering

Three purchase summaries form the compact representation. Purchase recency is the elapsed number of days from the last transaction event to the snapshot. Purchase occasion count is the number of distinct nonempty transaction identifiers. Purchase event count is the number of transaction rows. The latter two measures can differ when several transaction rows share an identifier. They describe purchase activity recorded in the event log and do not substitute for expenditure.

Nine additional variables expand the compact representation into the rich representation. They describe activity volume, conversion, timing and category behaviour. Table 3.1 defines all twelve variables. The three purchase variables are carried into the rich table without recalculation, so each purchaser has the same purchase summary in both representations.

Table 3.1 Recorded feature contract for the two customer representations

Feature	Block	Operational definition
Purchase recency days	Compact transaction	Days from the latest transaction event to the snapshot
Purchase occasion count	Compact transaction	Distinct nonempty transaction identifiers
Purchase event count	Compact transaction	Transaction event rows
Total events	Activity	All event rows for the visitor
Active days	Activity	UTC calendar days containing an event
Cart per view	Conversion	Add to cart events divided by view events
Transaction per view	Conversion	Transaction events divided by view events

Feature	Block	Operational definition
Activity span days	Temporal engagement	Days between first and last event
Mean interevent hours	Temporal engagement	Mean gap between adjacent events
Session count 30m	Temporal engagement	Sessions separated by more than thirty minutes
Distinct top categories	Category behaviour	Distinct root categories among known joins
Top category share	Category behaviour	Largest root category count divided by known category events

Category variables require a time-valid join because item properties change during the observation period. Each event receives the latest category property that was available at or before its timestamp. The category tree then maps that category to its root. This procedure covers 2,099,173 event rows, which is 76.16 per cent of all events. Unknown joins are excluded from category denominators rather than treated as a category. A total of 1,168 purchasers have no event with a known category. Their two category features are set to zero, while a separate coverage indicator records the missing information. This indicator is used for data checks and is not included among the twelve clustering variables. The missing category records prevent these variables from being interpreted as complete purchase preferences.

Ratio variables also use explicit denominator rules. Cart per view and transaction per view equal zero when a visitor has no recorded view event. This condition affects 428 purchasers and is retained in the feature audit. A session starts with a visitor's first event or after an inactivity gap greater than thirty minutes. Alternative gaps of fifteen and sixty minutes are reserved for sensitivity analysis. These choices make every derived value reproducible while exposing the assumptions most likely to affect temporal structure.

Nonnegative counts and time variables receive a logarithm after adding one. Ratios and shares remain on their original scale. Each representation is then standardised feature by feature using its own fitted means and standard deviations. Full-data clustering fits these transformations on the complete cohort. Sample splitting and resampling fit them only on the relevant training or subsample rows. PCA is applied only to a separately standardised copy of the rich representation and never supplies features to rule explanations or cluster descriptions.

Before clustering, both feature tables were checked against the source data. File hashes identify the exact raw inputs. Unit tests recalculate features for small manually checked

event histories that include equal timestamps, category changes, missing categories and zero denominators. Each table must contain the same 11,719 unique purchaser identifiers in the same order. The three purchase variables are then compared row by row and must be identical across the two tables. These checks ensure that the compact and rich analyses differ through the nine additional behavioural variables rather than through different customers or conflicting purchase summaries.

3.2 Analytical methods and mathematical scope

The comparison uses one recorded customer cohort, feature contract, set of candidate grids, deployment constraints and evaluation measures. K Means provides a centre-based reference partition that can be approximated by ExKMC rules. CLASSIX provides a separate clustering result with native aggregation provenance. ExKMC is not a clustering algorithm, and no composite score ranks it against CLASSIX. The same change in represented information is traced through each partition and its corresponding explanation. Before any hyperparameter grid is given, this section first explains what each clustering method mathematically computes, because a grid of settings is only meaningful once the underlying optimisation and its dependence on representation are understood.

3.2.1 What K Means computes

K Means partitions the standardised customer vectors $x_1, \dots, x_n \in \mathbb{R}^d$ ($n = 11,719$; $d = 3$ for the compact representation or $d = 12$ for the rich representation) into k disjoint clusters. Writing $C_1, \dots, C_k$ for the clusters and $\mu_1, \dots, \mu_k \in \mathbb{R}^d$ for their centroids, K Means seeks the assignment and centroids that minimise the within-cluster sum of squared Euclidean distances,

$$\min_{C_1, \dots, C_k, \mu_1, \dots, \mu_k} \sum_{j=1}^k \sum_{i \in C_j} \|x_i - \mu_j\|_2^2.$$

This is the objective originally proposed by MacQueen (1967). Given a fixed assignment, the centroid μ_j that minimises this objective for cluster j is simply the coordinate-wise mean of the points assigned to it. Given fixed centroids, the assignment that minimises the objective places each point in the cluster whose centroid is nearest in Euclidean distance. This second property means K Means always induces a *Voronoi partition* of the feature space: the region belonging to centroid μ_j is exactly the set of points closer to μ_j than to any other centroid, and these regions are convex polyhedral cells whose boundaries are the perpendicular bisectors between pairs of centroids. K Means is therefore a *centroid-based* method operating entirely on Euclidean geometry: two customers are

judged similar only insofar as their standardised feature vectors are numerically close under the ordinary Euclidean distance.

Minimising this objective exactly is NP-hard in general, so K Means alternates between the two steps above (reassign points to their nearest current centroid, then recompute centroids as cluster means) until neither step changes the partition. This procedure only ever reaches a *local* optimum, and which local optimum it reaches depends on where the centroids started. *k-means++* is a seeding rule that chooses the initial centroids sequentially, picking each new centroid at random with probability proportional to its squared distance from the nearest centroid already chosen, so that the initial centroids are spread across the data rather than clustered together (Arthur and Vassilvitskii, 2007); this improves the expected quality of the local optimum reached and speeds convergence, but it does not guarantee that the global optimum is found. *Multiple initialisations* — refitting from several independent *k-means++* seeds and keeping the run with the lowest objective value — further reduce, without eliminating, the risk of settling on a poor local optimum. This dissertation uses *k-means++* seeding, twenty initialisations and the fixed seed 20260801 for every K Means fit (MacQueen, 1967; Arthur and Vassilvitskii, 2007), so that the residual variation across repeated fits with the same data and parameters is attributable only to the finite number of initialisations, not to an unfixed source of randomness.

Because the objective sums *squared* per-dimension differences, K Means is directly sensitive to feature scaling: a feature recorded on a larger numeric scale, or with larger variance, can dominate the Euclidean distance purely because of its units, independently of how informative it actually is. This is why every feature, in both representations, is standardised to zero mean and unit variance before clustering (§3.1.2) — standardisation gives each of the three or twelve features comparable influence on the distance calculation, rather than letting the raw units of, say, a day count and a ratio determine their relative importance by accident. This dependence on scaling is also the direct mechanical link between representation and outcome that motivates this dissertation’s central comparison: adding nine standardised behavioural dimensions to the three purchase dimensions changes *every* pairwise Euclidean distance between customers, because Euclidean distance is computed jointly over all d coordinates. A different set of pairwise distances can move where the objective-minimising centroids fall, and can therefore change which Voronoi cell — and so which final cluster — a given customer falls into, even though that customer’s three original purchase coordinates are numerically identical in both the compact and the rich fit. Chapter 4 traces this mechanical link through to the observed low cross-representation agreement (RQ1).

3.2.2 What CLASSIX computes

CLASSIX (*fast and explainable clustering via sorting*) reaches a partition by a different route from K Means, and is introduced here in the stages it actually executes, before any of its hyperparameters are discussed (Chen and Güttel, 2024).

1. **Standardised observations.** As with K Means, CLASSIX operates on the same standardised customer vectors $x_i \in \mathbb{R}^d$ used throughout this dissertation.
2. **Principal sorting direction.** CLASSIX computes a single unit vector $u \in \mathbb{R}^d$, the *principal direction*: the direction along which the standardised data have maximum variance, equivalently the leading eigenvector of the data’s covariance matrix (the same direction principal component analysis would select as its first component; §3.2.6 below uses this same covariance matrix and its eigenvalues directly).
3. **Scalar projection.** Every observation is projected onto that one direction, giving a single number $s_i = u^\top x_i$, its *projected score*. This collapses a d -dimensional point onto a one-dimensional line while preserving, approximately, its position along the direction of greatest spread in the data.
4. **Sorting observations.** All n customers are sorted into ascending order of their projected score s_i . Customers with similar scores — and therefore, typically, similar positions along the principal direction — become neighbours in this sorted order.
5. **Aggregation around starting points.** CLASSIX walks through the sorted order. Whenever the current point does not lie within a fixed *aggregation radius* r (measured by ordinary Euclidean distance, in the full d -dimensional space) of the current group’s *starting point* — the first, representative point of that group — a new group is started, with the current point becoming the new starting point. §3.2.5 below shows why the one-dimensional sorted score can be used to rule out many of these full-distance checks in advance.
6. **Aggregation groups.** The result of this walk is a collection of *aggregation groups*: each group is a starting point together with every subsequent point (in sorted order) found to lie within radius r of it. A dataset of n points can produce anywhere from one aggregation group (if all points are mutually close) up to many small groups (if points are spread out); this dissertation’s fitted customer data produce 131 groups in the compact representation and 559 in the rich representation (§4.3).
7. **Merging groups.** Because aggregation only ever compares a point with its own group’s starting point, two aggregation groups whose starting points are themselves close together are then merged into one cluster. CLASSIX offers two merge rules: *distance merging*, which merges two groups only when the Euclidean distance between their starting points is below a merge threshold (the aggregation radius scaled by a configured merge factor), and *density merging*, which instead merges

two groups when the local density of observations found between their two starting points exceeds a threshold — a criterion that can join two groups whose starting points are further apart than the distance-merging threshold would allow, provided enough intervening points bridge them. §3.2.6 below derives a stability condition that applies only to the distance-merging branch.

8. **Minimum-size and noise handling.** After merging, any resulting cluster (or any aggregation group left unmerged) that is smaller than a configured minimum size is not accepted as a valid cluster. Depending on configuration, its member points are either folded into a larger neighbouring cluster (a *tiny group merge*) or, if not, labelled *noise*: customers who are not assigned to any final cluster because no aggregation group containing them met the minimum-size or merging conditions required to become one. A noise label is itself a specific, recorded outcome of the aggregation and merging process, not a missing value or a data-quality problem, and — as used throughout Chapters 3 and 4 — a noise assignment counts as a difference in outcome whenever two partitions are compared.
9. **Final cluster assignments.** Every customer’s final label is either one of the merged cluster identifiers or the noise label.
10. **Provenance.** Because every one of the steps above is recorded — which aggregation group a customer’s point fell into, that group’s starting point, and the sequence of merge decisions that connected the group to its final cluster (or determined that it remained noise) — a customer’s final label can always be traced back through this explicit chain. This traceable, step-by-step record is what this dissertation calls CLASSIX’s *provenance*: it is a property of how the partition was *computed*, not a separately fitted description of the partition after the fact (contrast with ExKMC’s post hoc trees, §3.2.4 and §4.3).

This staged process explains why CLASSIX is also sensitive to representation, through a different mechanism from K Means: adding nine standardised dimensions changes the covariance matrix used to compute the principal direction u (stage 2), which changes every customer’s projected score s_i (stage 3), which changes the sorted order (stage 4) and therefore which points are examined together during aggregation (stage 5) — all before the full Euclidean aggregation radius is ever applied. A different principal direction is one concrete, mechanistic reason two representations of the same customers can produce different CLASSIX aggregation groups and, ultimately, different final clusters.

3.2.3 Clustering candidates and selection

K Means candidates range over $k = 2$ to 12, using the seeding, initialisation and objective described above. CLASSIX candidates use the aggregation and merging process described above with Euclidean distance and sorting along the first principal direction; the

aggregation radius r ranges from 0.25 to 1.75 in increments of 0.05, the minimum cluster size takes values 1, 5 and 10, and group merging uses either the distance or the density rule defined above. The merge scale is 1.5, tiny groups can be merged into a neighbouring cluster, and post allocation (reassigning noise points after the main fit) is disabled. Each representation has 11 K Means candidates and 186 CLASSIX candidates. A recorded boundary diagnostic triggered a symmetric radius extension from 1.80 to 2.25 for both representations; the extension tests grid sensitivity and does not replace the primary selection.

Each method and representation reports an unconstrained silhouette optimum and a deployment-acceptable result. *Silhouette* compares, for each point, its average distance to other points in its own cluster against its average distance to points in the nearest other cluster; it is higher when clusters are internally compact and well separated from each other, and its value lies in $[-1, 1]$ (Rousseeuw, 1987). *Davies–Bouldin* averages, over all clusters, the ratio of within-cluster scatter to between-cluster centroid separation, so *lower* values indicate more compact, more separated clusters (Davies and Bouldin, 1979). *Calinski–Harabasz* is the ratio of between-cluster to within-cluster variance (scaled by degrees of freedom), so *higher* values indicate more separation relative to internal spread (Caliński and Harabasz, 1974). A candidate is internally valid only when at least two nonnoise clusters remain and all three indices are finite. Deployment acceptability further requires 2 to 12 nonnoise clusters, no more than 20 per cent noise and no single nonnoise cluster larger than 95 per cent of nonnoise observations. Selection maximises silhouette, then prefers lower Davies–Bouldin, higher Calinski–Harabasz, a smaller largest cluster share, less noise, fewer clusters and finally the lexical parameter record. Runtime is measured but never selects a model. Keeping the unconstrained result reveals when an attractive internal score is produced by a partition that is unsuitable under the stated operational constraints — for example, a partition that places nearly every customer in one cluster and a handful of outliers in a second (§4.2).

Internal indices exclude observations labelled as noise, so CLASSIX index comparisons refer to each partition’s nonnoise subset. Cross-representation partition comparisons retain every purchaser because a noise assignment is itself a difference in assignment. Fixed K comparisons use the same value from 2 to 12. Fixed CLASSIX comparisons use the same radius, minimum size and merge rule. Independently selected deployment partitions are compared using three partition-agreement statistics. The *adjusted Rand index* (ARI) counts, over all pairs of customers, whether the two partitions agree on whether the pair is in the same cluster or different clusters, then corrects this count so that a random relabelling of either partition scores approximately 0 while identical partitions score 1 (Hubert and Arabie, 1985). *Normalised mutual information* (NMI) measures how much information one partition’s labels carry about the other’s, expressed on a

0-to-1 information-theoretic scale and normalised here by the arithmetic mean of each partition’s own entropy, so the statistic does not automatically reward a larger number of clusters. *Variation of information* (VI) is a complementary information-theoretic *distance* between two partitions, reported in nats (natural-log units, because entropy is computed with the natural logarithm here): a value of 0 means the partitions are identical, and larger values mean less shared information (Meilă, 2007). Because cluster labels are arbitrary integers assigned independently within each fit, a *Hungarian match* is used before building transition tables: this is an assignment algorithm that finds the one-to-one correspondence between two partitions’ labels that maximises total overlap, so that ”cluster 2 in the compact fit” can be meaningfully compared with ”cluster 2 in the rich fit” (Kuhn, 1955). The resulting migration share describes changed assignments under that matched correspondence and does not imply that numeric labels identify persistent customer types across representations.

Business *addressability* is evaluated for selected CLASSIX and K Means clusters through a documented audit rather than subjective cluster names. A nonnoise cluster must contain at least 50 customers and one per cent of nonnoise customers, retain at least 80 per cent category coverage, and differ from the global median by at least half an interquartile range on two to four named features. Passing this audit shows only that a cluster can be expressed as a limited set of query conditions with adequate coverage; it does not demonstrate intervention effectiveness, revenue change or human comprehension (§4.3, §4.3.1).

3.2.4 Resampling and explanation evaluation

Fixed-parameter overlap-resampling *stability* uses ten pairs of independent 80 per cent samples without replacement. The same seed pairs are used for both representations and methods. Each sample refits its transformations and clustering with the full-data deployment parameters. Adjusted Rand index is calculated only on purchaser identifiers present in both members of a pair. The statistic measures sensitivity to sampled customers and, for K Means, initialisation, *conditional on the selected configuration* — that is, it measures whether the same fixed recipe reproduces a similar partition when applied to a slightly different 80 per cent of customers. It does not measure stability of model *selection* (whether the same configuration would be chosen again from the full grid), calendar-time stability, or stability on an external cohort.

The CLASSIX explanation retains public aggregation group assignments, starting point indices, final labels and cluster sizes, in the sense defined in §3.2.2 above. Merge membership is reconstructed from public group and final assignments because version 1.5.1 does not expose a documented complete edge list. Every exported provenance

label is checked against the selected main result for the same visitor. The trace is an explanation of the fitted CLASSIX partition and does not claim that a group boundary is a human-validated customer rule.

ExKMC approximates K Means labels under five deterministic 80 per cent training and 20 per cent test splits, as a post hoc decision tree fitted *after* the K Means partition already exists (§2.3). The primary full-cohort selection gives four K Means clusters in both representations, so K is fixed at four before splitting. Log transformations, standardisation, the K Means reference and the ExKMC tree are fitted using training rows only. The test target is the label returned by the fitted K Means predictor. This estimates explanation fidelity conditional on the primary K choice rather than nested selection of K . The *leaf budget* — the maximum number of terminal nodes (leaves) the tree is allowed to grow to, with each leaf corresponding to one exported rule — ranges from K to four times K . The main comparison fixes the budget at K , while the wider range measures the fidelity and complexity tradeoff. A supplementary analysis selects K from 2 to 12 using each training split alone. Fidelity is the proportion of exact label agreements between the tree and the held-out K Means labels. Complexity is reported through effective leaves, depth, path conditions and features. Rule thresholds are exported in both standardised and original units.

3.2.5 Projection filtering and complexity

Every candidate CLASSIX fit must decide, for a potentially large number of pairs of observations, whether one lies within the aggregation radius of another (stage 5 of §3.2.2 above). Checking every pair by computing the full d -dimensional Euclidean distance would be correct but wasteful, because CLASSIX has already sorted every observation by its one-dimensional projected score before this step is reached. This section states and proves *why* that one-dimensional sorted order can be used to rule out some full-distance comparisons in advance, without ever discarding a comparison that the exact Euclidean check would have accepted: sorting observations by their projection allows some Euclidean distance checks to be ruled out before the full distance is ever calculated, because a large enough gap between two projected scores already guarantees that the full Euclidean distance must exceed the aggregation radius. This matters for the dissertation’s argument because it clarifies exactly what computational mechanism produces CLASSIX’s aggregation groups and, through them, its provenance — a mechanism structurally different from ExKMC’s post hoc tree fitting, and one whose computational cost is worth stating precisely rather than assumed to be linear.

Let processed (standardised) observations be $x_i \in \mathbb{R}^d$ ($d = 3$ or 12 , as above), let $u \in \mathbb{R}^d$ be the unit sorting direction from §3.2.2 (so $\|u\|_2 = 1$), and let $s_i = u^\top x_i$ be observation

i 's projected score. CLASSIX orders the scores and begins a group at a starting point c (itself one of the x_i , with score $s(c) = u^\top c$). Its aggregation radius, in the standardised coordinates used throughout this dissertation, is r .

Proposition 3.1. *If a later point (in sorted order) satisfies $s_i - s(c) > r$, then it cannot satisfy $\|x_i - c\|_2 \leq r$.*

Proof. By the Cauchy–Schwarz inequality, applied to the unit vector u and the difference vector $x_i - c$,

$$|s_i - s(c)| = |u^\top (x_i - c)| \leq \|u\|_2 \|x_i - c\|_2 = \|x_i - c\|_2,$$

using $\|u\|_2 = 1$. So the (one-dimensional) score gap is never larger than the (full d -dimensional) Euclidean distance. If the strict inequality $s_i - s(c) > r$ holds, the displayed bound forces $\|x_i - c\|_2 > r$ as well, so x_i cannot lie within the aggregation radius of c . The implementation includes equality in both the score window and the exact squared-distance comparison, so this filter never rejects a point the exact Euclidean check would have accepted (a mathematical false negative), apart from ordinary floating-point rounding. $\square$

In plain terms: if two observations are already separated by more than r along the one unit projection direction u , their full Euclidean distance cannot possibly be $\leq r$ either, because projecting onto u can only ever shrink a distance, never enlarge it. The later observation can therefore be safely ruled out of the current aggregation group using only its one-dimensional score, without ever computing the full d -dimensional distance. Sorting observations by their projected score groups points that are likely to be close together in the sorted order, so this rule lets CLASSIX skip the full-distance calculation for the (often many) pairs that are already far apart along the principal direction, while still checking the full Euclidean distance for every pair the rule cannot rule out.

This filter gives *exact* pruning for a fixed sorted order — it never changes which points end up in the same group compared with checking every full distance directly — but it does not make the resulting greedy group assignment independent of that order: two different valid sort orders of the same data could, in principle, walk through points in a different sequence and produce different starting points and groups. If n observations produce q aggregation groups, aggregation retains at most $n(n - 1)/2$ exact point comparisons, and distance merging retains at most $q(q - 1)/2$ starting-point comparisons. The worst-case distance arithmetic is therefore

$$O((n^2 + q^2)d)$$

in addition to principal-direction estimation and scalar sorting. It is important to distinguish three different things that might be meant by ”how much work CLASSIX does”: this $O((n^2 + q^2)d)$ bound is a *worst-case theoretical bound* on the number of exact distance comparisons that could be retained after filtering; the CLASSIX package’s own internal counter, separately, records the number of aggregation comparisons *actually* retained on the specific observed data (§4.3 reports 131 and 559 aggregation groups for the compact and rich fits); and the *empirical runtime* recorded in §4.1’s benchmark is a wall-clock measurement on one machine. None of these three quantities substitutes for either of the others, and in particular the package counter — an empirical quantity observed on this dataset — is not a proof that CLASSIX’s aggregation step runs in linear time in general; it only reports what happened on the data actually fitted.

Conditional determinism requires more than a fixed random seed. A positive leading eigengap (defined precisely in §3.2.6 below) fixes the one-dimensional principal subspace, while the implementation’s sign convention fixes its orientation once that convention’s defining quantity is separated from zero. The sorted order is fixed when relevant scores are distinct. Exact ties require an explicit stable secondary order or an order-invariance check. Radius, merging, minimum-size and nearest-eligible-group decisions must also avoid unresolved equalities. These conditions distinguish deterministic execution on the observed input from a global statement about all nearby inputs — a distinction taken up directly, for small perturbations of the input, in the next section.

3.2.6 A local stability certificate

Chapter 4’s sensitivity analysis (§4.4.2) perturbs the feature representation with small amounts of Gaussian noise and checks whether the fitted CLASSIX partition changes. Because CLASSIX begins by sorting observations along the principal direction u (§3.2.2, stages 2–4), one plausible source of instability under such a perturbation is that the perturbation itself rotates the principal direction, which would reorder the sorted scores and could, in turn, change which points are aggregated together. This section derives *sufficient conditions* under which that does not happen: conditions under which every finite decision that defines a distance-merging CLASSIX fit — the sorted order, the aggregation-window membership tests, and the merge-distance tests — remains exactly unchanged after a sufficiently small perturbation, so that the fitted partition itself is provably unchanged. This complements, but is distinct from, the empirical perturbation results reported in §4.4.2: the certificate below is a mathematical guarantee under stated conditions, whereas §4.4.2 reports what was actually observed when the perturbation was applied.

Let S be the processed (standardised) data's covariance matrix, with leading eigenvalues $\lambda_1 \geq \lambda_2$. Define the *eigengap* $\gamma = \lambda_1 - \lambda_2 > 0$: intuitively, a larger eigengap means the leading eigenvector (the principal direction u) is more sharply preferred over the second eigenvector, so a given amount of perturbation to the covariance matrix produces a smaller rotation of u — this intuition is made precise by the Davis–Kahan theorem below. Let a perturbation change the covariance matrix by ΔS , and let $\hat{u}$ denote the resulting perturbed principal direction, sign-aligned with the original u (so that $\hat{u} \rightarrow u$, rather than $\hat{u} \rightarrow -u$, as the perturbation shrinks to zero). The *Davis–Kahan theorem* bounds how far a matrix's leading eigenvector can rotate under a bounded perturbation of the matrix, as a function of the perturbation's size and the eigengap; it does not say anything about any of CLASSIX's subsequent aggregation, merging or noise decisions, only about this one geometric quantity, the direction u . When $2\|\Delta S\|_2 < \gamma$ (the perturbation is small relative to the eigengap), the theorem gives the conservative bound (Davis and Kahan, 1970)

$$\sin \angle(\hat{u}, u) \leq \frac{2\|\Delta S\|_2}{\gamma}.$$

In words: the sine of the angle between the original and perturbed principal directions is bounded by twice the perturbation's size, divided by the eigengap — so a larger eigengap directly buys a tighter bound on how much the sorting direction can rotate for a given perturbation. After sign alignment, this angular bound converts to a bound on the vectors' Euclidean distance, $\|\hat{u} - u\|_2 \leq \sqrt{2} \sin \angle(\hat{u}, u)$. If every processed point changes by at most ε_x under the perturbation and every processed point has Euclidean norm at most B , then the resulting change in any point's projected score is bounded by

$$b_s = \varepsilon_x + \sqrt{2}B \frac{2\|\Delta S\|_2}{\gamma}.$$

This score-change bound b_s combines two sources of movement: ε_x , the point's own direct displacement under the perturbation, and the second term, the extra score movement caused purely by the sorting direction itself rotating.

The remaining symbols state, for each of CLASSIX's finite decisions in turn, how large a safety margin that decision currently has against the perturbation. Let g_s be the minimum gap, in the sorted order, between the projected scores of adjacent observations (the sorted-order safety margin); let g_o be the absolute value of the score that the implementation's orientation (sign) convention is evaluated against (the sign-flip safety margin). Let g_w be the minimum margin, over every relevant pair, between a projected score difference and the aggregation-window boundary r (the aggregation-acceptance safety margin). Let g_r be the minimum margin between any tested full-distance aggregation check and r (the exact-distance safety margin). For the distance-merging branch specifically, let g_m be the corresponding minimum margin between any tested

starting-point distance and the active merge threshold (the merge-decision safety margin); let b_r and b_m bound how much those two distance tests can themselves change under the perturbation, analogously to b_s above.

Proposition 3.2. *The fitted partition under distance merging is unchanged by the perturbation if $g_s > 2b_s$, $g_o > b_s$, $g_w > 2b_s$, $g_r > b_r$ and $g_m > b_m$, provided that the representatives, eligible groups, minimum-size decisions, tiny-group rule, post-allocation rule and all relevant tie decisions remain fixed.*

Proof. The first inequality preserves every relevant pairwise score order because each endpoint of a compared pair can move by at most b_s , so a gap larger than $2b_s$ cannot be closed or reversed. The second preserves the package’s orientation sign, because the sign-defining quantity cannot cross zero if it starts more than b_s away from it. The third fixes window membership (whether a point falls inside or outside the aggregation radius, judged by score) because each relevant score difference can move by at most $2b_s$ (the sum of two endpoints’ movements). The fourth preserves every accepted or rejected full-distance aggregation comparison, since that comparison’s value cannot move past r if it started more than b_r away from it; the fifth preserves every merge comparison by the same argument applied to b_m and the merge threshold. With the branch (distance merging), the set of representatives, and the eligible comparison set all held fixed by these five conditions, every one of CLASSIX’s underlying finite decisions is unchanged, and unchanged finite decisions reproduce exactly the same aggregation groups, the same merge components, and therefore the same final labels. $\square$

This certificate must not be read as more than it is. It is *local*: it applies only within a neighbourhood of the original fit small enough that the stated margins exceed the stated bounds, not globally across arbitrary perturbations. It is *sufficient, not necessary*: a fit could still happen to be unchanged even when one of the five margin conditions fails, since the conditions are deliberately conservative. It is *conditional* on the branch and on every other finite decision (representatives, eligible groups, minimum size, tiny-group and post-allocation rules, tie-breaking) remaining fixed as stated. It applies *only to the distance-merging branch*: it does not cover density merging, whose merge decision depends on the local density of intervening points rather than on a single starting-point distance, and would require separately derived margins that this dissertation does not attempt. The Davis–Kahan bound itself controls only the rotation of the principal direction u ; it says nothing on its own about aggregation, merging, minimum-size or noise decisions unless the additional finite margins above are also shown to hold. Most importantly for this dissertation’s argument, this proposition is **not a general proof that CLASSIX is stable**, and it is **not a stability certificate for the rich-representation CLASSIX result reported in Chapter 4**: that selected configuration uses *density*

merging (§4.2, Table 4.2), which this certificate explicitly does not cover. The empirical Gaussian-perturbation results reported in §4.4.2, and the empirical customer-resampling results reported in §4.2, are therefore complementary to, but distinct from, this mathematical statement — one is a proof under stated sufficient conditions for one branch of the algorithm, the other is an observation of what actually happened when the fitted rich-representation (density-merging) partition was perturbed. Neither substitutes for the other.

3.2.7 Benchmark and robustness designs

Nine labelled datasets provide an algorithm-level benchmark for CLASSIX, K Means, and two further methods used only in this benchmark: DBSCAN, a density-based clustering algorithm that groups points found in dense, connected regions of the feature space and labels sparse points as noise, and Ward’s method, an agglomerative hierarchical clustering algorithm that repeatedly merges the pair of clusters that would cause the smallest increase in within-cluster variance (Ester et al., 1996; Ward, 1963). Candidate selection uses only internal indices and deployment guards, exactly as described in §3.2.3; the datasets’ reference labels are reserved for the subsequent ARI and NMI evaluation and never influence selection. K Means results average five fixed seeds before parameter selection. Ward, DBSCAN and CLASSIX store one structural row per candidate, while repeated fits audit label identity and runtime on the recorded environment. This empirical identity check is not a global determinism theorem.

The rich representation undergoes twelve leave-one-feature-out fits (one fit with each of the twelve named features removed in turn). PCA — principal component analysis, which re-expresses the standardised features as a smaller set of uncorrelated linear combinations ordered by the variance each explains, without using any cluster labels (Pearson, 1901; Hotelling, 1933) — retains at least 90 per cent of standardised variance and is compared with the named feature space on structure and stability only, never on rule quality, because its rotated components no longer correspond to the twelve named, business-meaningful variables. *CLASSIX perturbation* adds independent Gaussian noise, at a stated standard deviation, to every standardised feature value, refits CLASSIX with the selected parameters on the resulting perturbed data, and measures agreement (by ARI) between the perturbed and unperturbed fits; this dissertation uses standard deviations 0, 0.000001, 0.0001, 0.001 and 0.01, with ten independent noise seeds at every nonzero level. Session counts are rebuilt with inactivity gaps of 15, 30 and 60 minutes, in place of the primary 30-minute definition (§3.1.2). A negative control appends independently permuted rich-only dimensions to unchanged compact features: this demonstrates that added dimensionality and standardisation alone *can* alter geometry even when the added columns carry no customer-aligned information, but it is a single deterministic existence

demonstration rather than a variance decomposition, and it is not used to estimate what share of the compact-to-rich effect is attributable to dimensionality as such (§4.4.3). These diagnostics retain the primary selected configurations unless a section explicitly reports complete feature reselection.

Artificial intelligence tools assisted with code drafting, data processing, numerical checks and source discovery. The author reran the analyses from saved code, verified retained sources and checked reported values against recorded outputs. Reproducibility details are supplied with the accompanying materials.

Chapter 4

Results and Analysis

The results follow the evidence chain established in Chapter 3, and are organised to answer the five research questions from §1.3 in turn. §4.1 first checks the four clustering implementations and the label-free selection procedure on nine externally labelled datasets, before either is trusted on RetailRocket. §4.2 then compares the fitted compact and rich partitions and their resampling stability (RQ1, RQ2). §4.3 compares CLASSIX provenance, ExKMC fidelity and cluster addressability (RQ3, RQ4). §4.4 tests whether these findings survive feature deletion, PCA, numerical perturbation, alternative session windows and a permutation control (RQ5), before §4.5 states the validity limits that bound every claim in this chapter. Throughout, each subsection reports a result, states the mechanism that plausibly produces it, gives the interpretation that mechanism supports, and marks the limitation of that interpretation — results are not left to speak for themselves.

4.1 Algorithm benchmark

Nine labelled datasets provide an algorithm check before the customer analysis, using the label-free selection procedure and the internal indices defined in §3.2.3. Six shape datasets and three multivariate datasets were evaluated with K Means, CLASSIX, and, for this benchmark only, DBSCAN and Ward’s method (defined in §3.2.7). Selection used only internal indices and deployment guards; reference labels entered only the subsequent ARI and NMI calculation, after selection, and never influenced which candidate was chosen.

Result. No method dominates the benchmark. K Means has the highest mean ARI across the nine selected results at 0.597, followed by Ward at 0.575, DBSCAN at 0.509 and CLASSIX at 0.508. CLASSIX and Ward share a mean rank of 2.44, while K Means has 2.39 and DBSCAN has 2.72. These averages conceal marked dependence on shape:

CLASSIX recovers the Jain reference exactly but reaches only 0.264 ARI on R15; DBSCAN recovers Spiral exactly and performs best on Aggregation and Compound, yet its selected Wine candidate has no valid internal score; K Means is strongest on Seeds and Wine, while Ward is strongest on R15.

Mechanism and interpretation. Each method’s strengths track its own structural assumptions: CLASSIX’s radius-based aggregation suits compact, well-separated shapes like Jain but is a poor match for R15’s many overlapping Gaussian clusters; DBSCAN’s density criterion suits Spiral’s connected, non-convex arcs; K Means’s convex Voronoi cells (§3.2.1) suit the roughly spherical clusters in Seeds and Wine. No single method wins because no single geometric assumption fits every dataset, and this dependence on shape — not any one method’s general superiority — is the benchmark’s main finding.

Limitation. This pattern of successes and failures shows that the four implementations, the label-free selector, and the external evaluation code all behave sensibly across a deliberately varied set of known structures; it is a check on the analytical pipeline, not evidence about which method is preferable for RetailRocket customers, whose true cluster shape (if any) is unknown. The customer conclusion in §4.2 onward rests on paired evidence within one cohort, not on this benchmark’s algorithm ranking.

Table 4.1 Highest external ARI after label free selection on each benchmark dataset

Dataset	Method	ARI
Aggregation	DBSCAN	0.910
Compound	DBSCAN	0.826
Iris	CLASSIX and K Means	0.568
Jain	CLASSIX	1.000
Pathbased	CLASSIX	0.497
R15	Ward	0.804
Seeds	K Means	0.481
Spiral	DBSCAN	1.000
Wine	K Means	0.897

Observed runtime depends on implementation and dataset size. Median selected runtime is 0.011 seconds for CLASSIX, 0.064 seconds for K Means, 0.003 seconds for DBSCAN and 0.004 seconds for Ward on the recorded machine. These are single wall-clock measurements on one machine for small datasets, not the worst-case theoretical complexity derived in §3.2.5, and they do not establish a universal speed ordering between the four implementations.

4.2 Representation and customer cluster structure

This section answers RQ1 and RQ2 (§1.3): how much do independently selected K Means and CLASSIX partitions agree across the two representations, and does that agreement coexist with high within-representation stability?

Result. Changing the represented information alters the selected customer structure for both clustering methods. Table 4.2 reports the independently selected deployment-acceptable configurations. K Means selects four clusters in each representation, but the agreement between the two four-cluster partitions is only 0.014 ARI. CLASSIX selects two compact clusters and three rich nonnoise clusters with 218 additional noise assignments (customers CLASSIX did not place in any final cluster, as defined in §3.2.2). Their cross-representation agreement is 0.068 ARI. Recall from §1.5 that these ARI values compare the compact-space and rich-space partitions of the *same* customers to each other, not to any external ground truth; both values are close to the score two unrelated random partitions of the same customers would obtain by chance, even though the customer identifiers and algorithm families are unchanged.

Table 4.2 Deployment acceptable clustering results on the shared purchaser cohort. Noise records customers CLASSIX did not assign to any final cluster because no aggregation group containing them met the minimum-size or merging conditions to become one (§3.2.2); K Means never produces noise. Stability is mean fixed-parameter overlap-resampling ARI across ten resampled pairs (§3.2.4).

Representation	Method	Clusters	Noise	Largest share	Silhouette	Stability
Compact	CLASSIX	2	0	0.909	0.595	0.985
Rich	CLASSIX	3	218	0.898	0.291	0.965
Compact	K Means	4	0	0.664	0.560	0.988
Rich	K Means	4	0	0.579	0.351	0.995

Mechanism. §3.2.1 and §3.2.2 give the mechanical reason for this disagreement: both methods compute Euclidean distances (K Means directly; CLASSIX via its principal-direction projection) over *all* standardised dimensions at once, so adding nine standardised behavioural dimensions changes every pairwise distance between customers, even though each customer’s three original purchase coordinates are numerically unchanged. A different set of distances can move K Means’s optimal centroids to different locations, and can change the covariance structure that determines CLASSIX’s sorting direction — both of which can change the resulting partition without any change to the underlying customers.

The internal indices do not move in a single direction. Compact CLASSIX has silhouette 0.595, Davies Bouldin 1.057 and Calinski Harabasz 4285.5. The rich result has silhouette 0.291, Davies Bouldin 0.859 and Calinski Harabasz 1516.9. These CLASSIX values are

calculated on each partition’s nonnoise observations. Davies Bouldin favours the rich result, while silhouette and Calinski Harabasz favour the compact result. Compact K Means has silhouette 0.560 and Calinski Harabasz 10459.1, compared with 0.351 and 3923.4 in the rich space. Davies Bouldin rises from 0.775 to 1.040. Richer behaviour produces a different geometry whose apparent quality depends on the criterion.

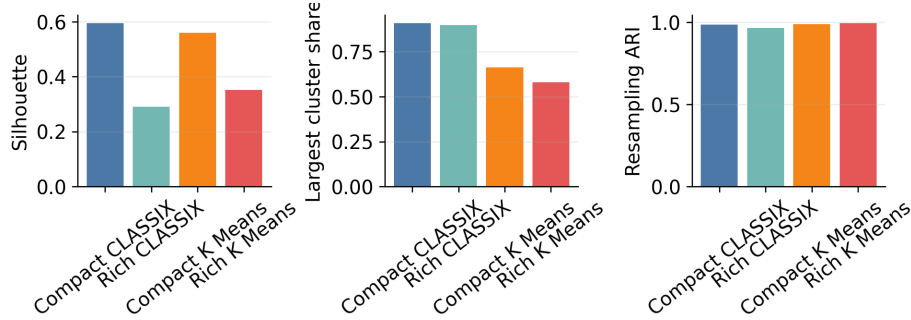


Figure 4.1 Internal structure and resampling stability for the selected customer partitions

The deployment rule materially changes the result that would be chosen by silhouette alone. Compact CLASSIX reaches silhouette 0.899 by placing 11,698 of 11,719 customers in one cluster. Rich CLASSIX reaches 0.792 with 11,716 customers in one cluster and the remaining three customers split across two tiny clusters. Compact K Means reaches 0.850 with 11,593 customers in one of two clusters. All three partitions fail the maximum share rule. The deployment selections sacrifice these high silhouette values to avoid reporting a nearly undivided customer base as a useful segmentation. Rich K Means is the only case where the unconstrained and deployment selections coincide.

Low agreement is not caused only by choosing different numbers of clusters. At every fixed K from 2 to 12, the K Means ARI between compact and rich representations remains between 0.012 and 0.113. At the common selected value of four it is 0.014, with NMI 0.089 and variation of information 1.691 nats. Under matched labels, 54.61 per cent of customers move to a different K Means cluster. Among 41 CLASSIX settings that are deployment acceptable in both spaces, matched-parameter ARI ranges from -0.051 to 0.392 with a median of -0.023 . The independently selected CLASSIX partitions have NMI 0.054 and variation of information 0.693 nats, with 18.66 per cent assigned to a different matched cluster or noise state. The lower CLASSIX migration share partly reflects the dominance of one large cluster in both representations and should not be read as stronger substantive agreement.

Cluster sizes make the changes more concrete. Compact K Means contains 7,784, 2,494, 1,344 and 97 customers. Rich K Means contains 6,790, 3,607, 1,156 and 166. Its 166 customer cluster has a median of 338.5 total events, 28 sessions and 12 top categories, whereas the full cohort medians are 6 events, 2 sessions and 1 top category. Another rich cluster of 1,156 customers has no known category at its median and a purchase recency

median of 127.4 days. These are descriptive contrasts in observed features, not customer identities or causal segments.

CLASSIX yields a sharper imbalance. The compact deployment result contains 10,656 and 1,063 customers. The rich result contains 10,331, 1,137 and 33 nonnoise customers plus 218 noise assignments. The 33-customer cluster is highly active, with medians of 869 total events, 36 sessions and 14 top categories. Its small size fails the documented addressability size rule. The 1,137-customer cluster has zero category coverage and fails the coverage rule. The dominant 10,331-customer cluster has no two sufficiently distinctive median effects. Richer features expose unusual behavioural combinations, but the selected CLASSIX partition does not turn them into a balanced set of directly addressable clusters.

Fixed-parameter overlap-resampling indicates that these structural differences are reproducible under the stated sampling scheme. Mean overlap ARI is 0.985 for compact CLASSIX and 0.965 for rich CLASSIX. The corresponding K Means values are 0.988 and 0.995. The lowest of ten pairwise values is 0.881 for compact CLASSIX and 0.941 for rich CLASSIX.

Interpretation. This is the answer to RQ2: high within-representation stability and low cross-representation agreement are not contradictory, because they answer different questions. Stability (§3.2.4) asks whether the *same fixed recipe* reproduces a similar partition when refit on a slightly different 80 per cent of customers; cross-representation ARI asks whether *two different recipes* (compact and rich) agree on the *same* customers. A configuration can be highly self-consistent — reliably rediscovering its own partition under resampling — while still describing a genuinely different grouping from the other representation’s equally self-consistent partition. This is direct evidence that representation is a modelling choice with its own reproducible consequences, not a preprocessing detail whose effect washes out once a method is ”properly” fitted.

Limitation. Stable refitting does not imply that a partition is uniquely correct, that model *selection* (choosing this configuration from the full grid in the first place) would recur under resampling, or that either partition would remain stable outside this cohort or over calendar time. It shows only that each selected representation and parameter combination tends to reproduce its own assignment pattern when one fifth of the cohort is omitted.

4.3 Explanation complexity, fidelity and addressability

This section answers RQ3 and RQ4 (§1.3): how does representation change CLASSIX’s intrinsic provenance and ExKMC’s post hoc fidelity, and which resulting clusters are operationally addressable? The explanation results separate provenance from

approximation throughout: CLASSIX exposes how its own observations are aggregated and merged (§3.2.2); ExKMC approximates a separately fitted K Means predictor with axis-aligned rules (§3.2.4). Their numbers answer different questions about different objects (§2.5) and are never combined into one explanation score.

Result (CLASSIX provenance). The selected compact CLASSIX fit produces 131 aggregation groups that merge into two final clusters. The rich fit produces 559 groups and four final states consisting of three clusters and noise. Every exported customer trace contains the visitor identifier, aggregation group, representative row and final cluster. Both exports reproduce the selected full data labels exactly with ARI 1, confirming the trace is a faithful record of the fitted partition rather than a separate approximation of it.

Mechanism and interpretation. Rich CLASSIX needs more than four times as many aggregation groups even though its final nonnoise cluster count rises only from two to three. This follows directly from the staged process in §3.2.2: with nine more standardised dimensions contributing to every pairwise distance, fewer customers fall within a fixed aggregation radius r of any one starting point, so more, smaller groups are needed before the merge step consolidates them into clusters. Group count is therefore a measure of local geometric granularity and computational provenance, not of how difficult the resulting clusters are for a person to read or act on — a large group count does not imply a confusing explanation, only a more finely subdivided intermediate trace.

Result (ExKMC fidelity). The main ExKMC comparison fixes K at four for both representations because both primary K Means selections contain four clusters. At the four-leaf budget, both trees use four effective rules with depth 3 and 2.25 conditions per rule. Compact mean training fidelity is 0.9466 and mean held-out fidelity is 0.9448. Rich fidelity is higher at 0.9639 and 0.9650. Across the five splits, compact held-out fidelity ranges from 0.9416 to 0.9480 and rich fidelity ranges from 0.9599 to 0.9697. This is descriptive split variability rather than inferential uncertainty. The close training and test values give no sign of a training-only gain that disappears on held-out customers.

Compact rules use purchase occasion count and purchase recency across all five splits. Rich rules use purchase occasion count, activity span and the number of distinct top categories. Thresholds are exported in original units after reversing the training scaler and logarithmic transform. Named thresholds support direct database style conditions, but they inherit the meaning and missingness limits of the constructed variables.

Additional leaves change the two fidelity frontiers differently. At sixteen leaves, compact mean held out fidelity reaches 0.9924 with mean depth 6.6 and 4.84 conditions per rule. Rich fidelity begins higher at four leaves but reaches 0.9747 with mean depth 6.4 and 4.74 conditions at sixteen leaves. Compact fidelity begins lower and gains more from expansion. A supplementary training only selection chooses two compact clusters and

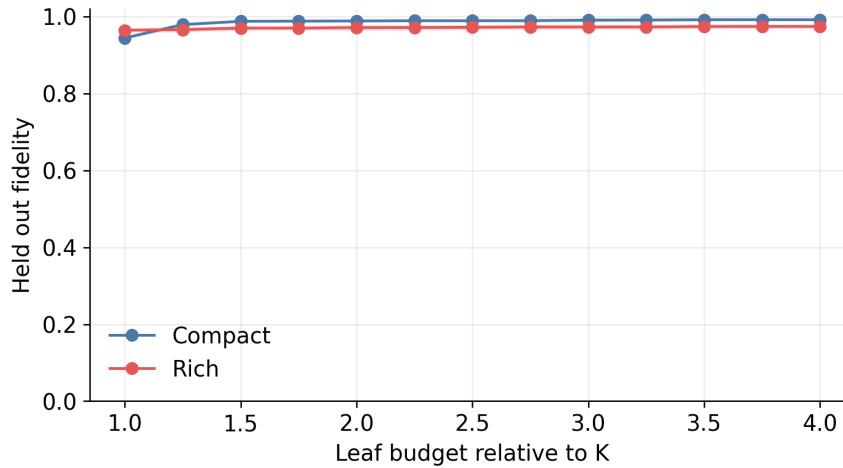


Figure 4.2 Held out ExKMC fidelity across the fixed leaf budget

four rich clusters in every split. Its compact fidelity is 0.9985 with two rules, but that result is not used for the main complexity comparison because K differs.

Interpretation. The crossover between the two fidelity curves — rich ahead at a four-leaf budget, compact ahead by sixteen leaves — shows that representation changes not only *whether* a small tree can approximate K Means well, but the entire *shape* of the complexity/fidelity tradeoff, not merely a single headline number.

Limitation. A higher ExKMC fidelity is evidence that the tree agrees more often with its own K Means reference; per §2.4, it is not evidence that the underlying K Means partition is itself more correct, more useful, or more stable, since fidelity is defined entirely relative to that partition.

4.3.1 Addressability: a stricter, distinct criterion

The preceding results show that a cluster can exist, be internally well separated (high silhouette or Calinski–Harabasz), and be highly reproducible under resampling (§4.2), while still failing to be usable as a targeted business segment. This dissertation therefore treats *addressability* (§3.2.3) as the last, and strictest, link in a chain of increasingly demanding properties: a partition can **exist** (any candidate that runs to completion); be **internally separated** (pass the silhouette/Davies–Bouldin/Calinski–Harabasz validity checks); be **stable** (survive fixed-parameter overlap-resampling, §4.2); be **rule-expressible** (approximated by a short ExKMC tree, or traced by CLASSIX provenance, above); be **addressable** (pass the size, coverage and distinctiveness audit); and, beyond anything measured in this dissertation, have **realised business value** (demonstrated to change an intervention outcome). Each property is necessary for the next but not sufficient for it, and this dissertation’s evidence stops at addressability.

Result. The addressability audit is stricter than rule export, and most selected clusters fail it despite having already passed the earlier, weaker checks. Neither compact CLASSIX cluster passes because neither differs from the cohort median on at least two named features by the required half interquartile range. In the rich result, the 33-customer cluster is too small, the 1,137-customer cluster has zero category coverage, and the 10,331-customer cluster lacks two sufficiently distinctive median conditions. None of the five nonnoise CLASSIX clusters is classified as addressable. No compact K Means cluster passes. One of four rich K Means clusters passes all gates, while the others fail because they have too many distinguishing conditions or insufficient category coverage.

Interpretation. The compact CLASSIX partition (Table 4.2) has the highest silhouette of any selected configuration in this dissertation, yet zero of its clusters are addressable; the rich K Means partition has the lowest compact-comparison silhouette among the four selected configurations, yet it is the only one to produce an addressable cluster. Internal validity and addressability are evidently not the same property and do not even rank configurations in the same order — a result that could not be seen without evaluating both explicitly.

Limitation. Business interpretation remains narrow at every stage of this hierarchy. Rich features expose a high-activity minority and a group with missing category histories, described here only as observed contrasts in recorded behaviour. ExKMC shows that K Means assignments *can* be represented with a small number of named thresholds at high held-out fidelity; it does not show that a person will find those thresholds meaningful. The one passing rich cluster meets a technical audit rather than an outcome test. The observed value in this dissertation is limited to queryability and auditable coverage — the fifth link in the chain above. Revenue effect, campaign response and human comprehension, the sixth link, remain entirely unmeasured and are not claimed.

4.4 Sensitivity and robustness

This section answers RQ5 (§1.3): do the representation effects reported in §4.2–§4.3 survive specific, named alternative explanations, or are they artefacts of one parameter choice? Each diagnostic below targets one such alternative explanation and is read as ruling that specific alternative in or out — none of them, individually or together, is a full variance decomposition of the compact-to-rich effect. Fixed checks (feature deletion, PCA, perturbation, session windows) measure local dependence around the already-selected configuration. Reselection repeats the candidate search and deployment rule from scratch after changing the features. The permutation control tests what happens when added dimensions carry no customer-aligned information at all.

4.4.1 Named features and reduced geometry

Result. Fixed parameter deletion identifies category behaviour as the strongest local influence on the rich partitions. Removing distinct top categories reduces agreement with the full rich result to ARI 0.189 for CLASSIX and 0.516 for K Means. Removing top category share gives ARI 0.238 and 0.517. Every other CLASSIX deletion remains between 0.927 and 0.967. The corresponding K Means values remain between 0.855 and 0.991. Activity span and mean interevent time have the next largest effect on K Means, with ARI 0.874 and 0.855.

Interpretation. This rules out one specific alternative explanation of the rich-representation effect: that it is spread roughly evenly across all nine added variables. Instead, influence is concentrated: removing either of the two category variables alone moves the rich CLASSIX partition roughly as far from its own full-feature result as the two representations are from each other cross-representation (§4.2), while removing any of the other seven added variables leaves CLASSIX largely unchanged. Category information is therefore a specific, identifiable driver of the observed rich-representation structure, not an unattributable side effect of adding "more dimensions" in general.

Limitation. Influence is unevenly distributed across the named variables, but this does not mean category information alone would generate a *preferable* segmentation, nor does removing one variable at a time capture interaction effects between variables removed together.

The PCA diagnostic preserves 91.32 per cent of standardised rich variance in six components. It is a fixed-parameter local diagnostic rather than PCA-space model selection. K Means in this reduced space agrees with the named rich partition at ARI 0.995 and has silhouette 0.390 compared with 0.351. CLASSIX retains three clusters, lowers noise from 218 to 71 customers and agrees with the named result at ARI 0.920. Its silhouette is 0.304 compared with 0.291. Mean resampling ARI is 0.995 for reduced K Means and 0.987 for reduced CLASSIX. The principal component space preserves most fitted structure. It is not used for explanation because rotated components do not retain the direct business meaning of the twelve named variables.

Complete reselection confirms that category information creates the largest structural turn. All twenty two scenarios produce a deployment acceptable candidate. Removing distinct top categories changes CLASSIX to eleven clusters with ARI 0.153 against the full rich partition. K Means changes to two clusters with ARI 0.728. Removing the whole category block produces seven CLASSIX clusters with ARI 0.126 and two K Means clusters with ARI 0.729. The progressive endpoints reproduce the compact and rich primary partitions exactly. These 4,334 candidate fits show that the category effect persists when parameters and cluster counts are allowed to change.

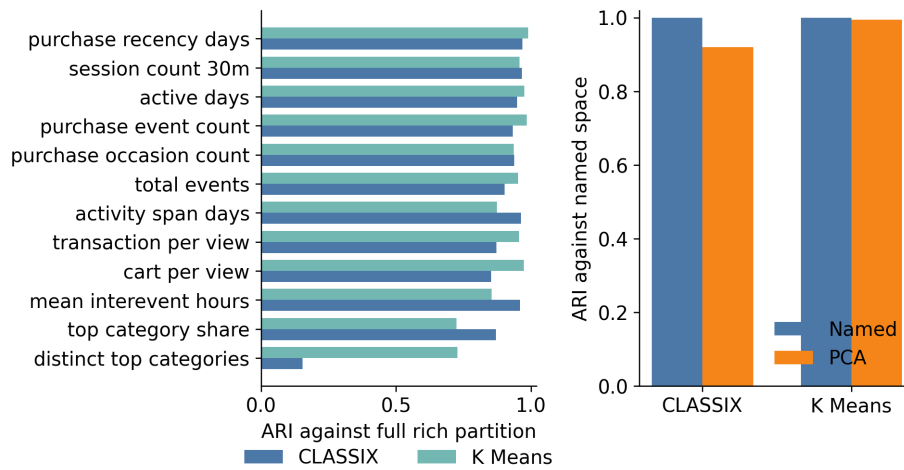


Figure 4.3 Reselected feature deletion and PCA diagnostics

CLASSIX sensitivity is distributed across the grid rather than concentrated at one radius. Deployment acceptable candidates occur at twenty three of thirty one compact radii from 0.35 to 1.45 and twenty rich radii from 0.30 to 1.25. Minimum size 1 produces no acceptable candidate in either representation. Minimum sizes 5 and 10 produce 23 and 32 acceptable compact candidates, compared with 19 and 27 rich candidates.

4.4.2 Perturbation and temporal assumptions

Result. Small Gaussian perturbations leave the selected CLASSIX partitions almost unchanged. Compact agreement remains ARI 1 through a noise standard deviation of 0.001. At 0.01 the mean ARI is 0.999 and the minimum is 0.996. Rich agreement is also exact through 0.0001. It falls to a mean of 0.995 at 0.001 and 0.985 at 0.01, with minima of 0.985 and 0.962.

Interpretation. This rules out a second specific alternative explanation: that the low cross-representation agreement reported in §4.2 is an artefact of numerical instability rather than a genuine geometric effect of the added variables. If the fitted partitions were fragile to tiny numerical noise, that fragility — not representation — could explain why compact and rich results differ. Instead, both partitions are essentially exact under noise far smaller than the perturbation needed to move any customer between clusters; the richer fit is measurably more sensitive at the largest tested noise level, consistent with §3.2.1–§3.2.2’s point that more dimensions contribute more terms to every distance calculation, but it does not become unstable in the sense of wholesale reassignment.

Limitation. These empirical results do not certify the density-merging branch used by the rich CLASSIX selection (Table 4.2); the mathematical certificate in §3.2.6 applies only to distance merging when its stated margins hold, and is a proof under stated conditions, not a description of what was empirically observed here.

Result. Reconstructing sessions from raw events gives a similar qualified result. The selected thirty minute window reproduces both partitions exactly. A fifteen minute window gives ARI 0.976 for K Means and 0.971 for CLASSIX. A sixty minute window gives ARI 0.985 and 0.959. The four-cluster K Means result and the three-cluster CLASSIX result persist at both alternative windows, although the CLASSIX noise count changes from 218 to 220 and 239.

Interpretation and limitation. This rules out a third alternative explanation: that the main findings depend on the specific choice of a thirty-minute inactivity gap. They do not, at the level of cluster count and overall structure. The detailed customer-level assignments and exact noise count, however, still depend on the session convention, so this robustness is qualified rather than absolute.

4.4.3 Dimensionality control

Result. The permutation control sets a stricter boundary on causal interpretation. The compact columns remain unchanged while nine additional dimensions, independently permuted across customers, are appended. These permuted columns preserve their own marginal distributions but destroy their joint relationship with the compact representation and with each other, by construction. With the compact K Means setting held fixed, adding all nine permuted columns lowers agreement with the compact partition to ARI 0.017 and silhouette to 0.127. With the compact CLASSIX setting held fixed, adding even one permuted dimension produces a single cluster and ARI 0.

Interpretation. This demonstrates that added dimensionality and standardisation *can*, by themselves, substantially alter both methods' fitted geometry, even for columns that are constructed to carry no genuine information about any customer — direct evidence for the scaling-sensitivity mechanism described in §3.2.1.

Limitation. This is a single deterministic existence demonstration, not a variance decomposition, and it must not be read as one. It shows *that* dimensionality alone can move geometry; it does not estimate *how much* of the observed, genuine compact-to-rich difference (§4.2) is attributable to dimensionality as opposed to the specific behavioural content of the nine real added variables. Named feature deletion (§4.4.1) and PCA (above) separately show that category and temporal information specifically, not dimensionality in the abstract, contribute reproducible structure in the observed rich representation — so the representation effect reported throughout this dissertation should be attributed to what the added variables measure, not merely to how many of them there are.

4.5 Integrated discussion and validity limits

The study concerns purchasers recorded in one anonymous event log. It excludes visitors without a transaction event and has no demographic, price, revenue, campaign or outcome data. Category features cover 76.16 per cent of events, while 1,168 purchasers have no known category. Zero ratios can indicate a genuine absence of views or the limits of recorded behaviour. These boundaries prevent substantive labels such as valuable, loyal or profitable from being inferred from the clusters.

Internal indices and fixed-parameter overlap-resampling measure geometric separation and conditional repeatability within the observed design. They do not establish an objectively correct number of customer groups or stability of complete model selection. The deployment constraints prevent nearly single-cluster solutions from dominating selection, but the thresholds are operational choices rather than universal standards. The nine labelled benchmarks provide algorithm checks across several shapes and dimensions, not a guarantee that their relative performance transfers to customer data.

Explanation evidence has separate limits. CLASSIX provenance is exact for its own fitted labels but does not produce a short business rule. ExKMC fidelity measures agreement with a K Means predictor and does not validate that predictor. The addressability hierarchy introduced in §4.3.1 checks size, coverage and observable distinctiveness, yet none of the selected CLASSIX clusters passes it, and only one of eight selected K Means/CLASSIX clusters overall does. Business value here means the capacity to inspect provenance, export named conditions and audit coverage — the fifth link in that hierarchy. Intervention response, commercial return and human understanding, its sixth link, require prospective evaluation beyond the available data and are not claimed anywhere in this dissertation.

Chapter 5

Conclusion

This dissertation asked five research questions (§1.3) about how changing the information used to represent the same 11,719 purchasers affects observable and explainable cluster structure. On RQ1 and RQ2, compact and rich representations produce substantially different assignments for both K Means and CLASSIX: the two representations' independently selected partitions agree with each other at an adjusted Rand index close to what unrelated random partitions of the same customers would score, whether or not the number of clusters happens to match, and this holds even though no external ground truth exists to say either partition is "correct" (§1.5). The compact space gives higher silhouette values, while the rich space gives a more balanced K Means partition and many more CLASSIX aggregation groups. Matched-parameter CLASSIX settings also give low or variable agreement. Fixed-parameter overlap-resampling remains high within each representation, so conditional repeatability — a configuration reliably reproducing its own partition — does not imply agreement across representations or stability of model selection; these are answers to different questions (§4.2).

On RQ3 and RQ4, the explanation results reveal a representation-dependent frontier. At four leaves, compact ExKMC fidelity is 0.9448 and rich fidelity is 0.9650 with equal tree complexity. At sixteen leaves, compact fidelity rises to 0.9924 while rich fidelity reaches 0.9747. CLASSIX provenance remains exact, yet the rich fit requires 559 aggregation groups compared with 131 — a difference in computational granularity, not necessarily in human interpretability (§4.3). Of eight selected clusters across both methods and representations, only one rich K Means cluster passes the addressability audit; the remaining seven fail on size, category coverage, or distinctiveness despite several having passed every weaker check (internal validity, resampling stability) that precedes addressability in the hierarchy set out in §4.3.1. Richer behaviour increases descriptive reach without guaranteeing simpler explanations, superior internal scores or deployment-ready groups.

On RQ5, the sensitivity programme (§4.4) rules out several specific alternative explanations of these findings — numerical fragility, the exact session-window choice, and dimensionality as such — while identifying category behaviour as the single largest, specifically attributable driver of the rich-representation effect.

Taken together, this evidence supports one bounded claim: representation is not a *neutral input* to clustering (§2.1) for this cohort, this feature contract, and these two algorithms. It determines the geometry against which clustering quality, stability and explainability are all judged, and changing it changes conclusions that might otherwise be attributed to the algorithm or the customers alone. This claim does not extend further than that: the benchmark (§4.1) establishes only that the implementations and metric path behave sensibly across varied labelled structures, not that any one algorithm is generally superior; the local stability certificate (§3.2.6) states only the margins sufficient for the distance-merging branch, not a general stability guarantee for CLASSIX; and no result in this dissertation measures whether representation-dependent clusters would respond differently to an actual business intervention. Future work should test whether such clusters predict prospective responses under complete category and value data, and whether analysts understand and act on the exported provenance and rules. Until then, technical queryability is not realised business value.

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Appendix A

Technical Reproduction Map

The reported experimental results are reproduced from the distributed data with `python-msrc.reproduce`. The fixed sequence covers the benchmark, customer clustering, explanations, sensitivity analyses, summary tables and figures. The optional `--compile-pdf` flag also rebuilds the report. Environment versions and random seeds are recorded in `environment.yml` and `protocol/analysis_protocol.json`.

Primary clustering selections are stored in `results/retailrocket/selected_configurations.csv`. Customer level labels, transitions, profiles and resampling records remain separate so that every reported aggregate can be recalculated. ExKMC training selection, held out fidelity and exported rules are stored in distinct files. CLASSIX provenance includes an exact label audit against the selected result.

The sensitivity directory separates fixed parameter diagnostics from complete candidate reselection. PCA records that reduced components are not permitted for rules. The zero perturbation rows have ARI 1. Session windows are reconstructed from the distributed purchaser timestamp table. Permuted dimensions retain the compact columns exactly while destroying their joint customer alignment.

The distributed feature tables and purchaser event timestamps are sufficient for the reported experiments. The optional `--rebuild-features` flag reconstructs these tables when the four original RetailRocket files are available. The repository `README.md` records the execution sequence, output map and licence boundary.